AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I095*

Team Name: *SIX-SEVEN*

Domain:*Advanced Materials, Manufacturing and Industrial Innovation*

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Problem Statement

All the industries in the world aim for the MRO that is Maintenance, Repair and Operation. But each of them differs in the way they handle it. One common existing approach of MRO of machineries is using the Acoustic cameras in the industries to identify the leakages in pipelines, voltage leakage, fault in machines and etc... But the existing method is limited with only 2-D mapping which does not provide precise location of leakages. And this method does not help us in finding the faults in electrical machines. Also, the existing method requires human intervention.

Problem Understanding

- This problem is mainly faced by the large scale industries, and palces with densely packed pipelines.
- The main cause is due to the densely packed industrial environment ,making it difficult to locate the fault.
- Fault finding takes time which in turn stalls the production, this results in large financial loss for the stakeholders.

Existing Gaps & Challenges

- The current acoustic camera requires a human to operate it, this is not a efficient way to approach the problem. As in industries there might be dense pipelines which is difficult for the humans to decode.
- The current product in the market is only capable of producing 2D output, which does not give precise information in a densely packed area.
- The existing implementation only focus on the leaks in the pipelines and tubes, they totally forget to account for the faults in electrical and mechanical machine.

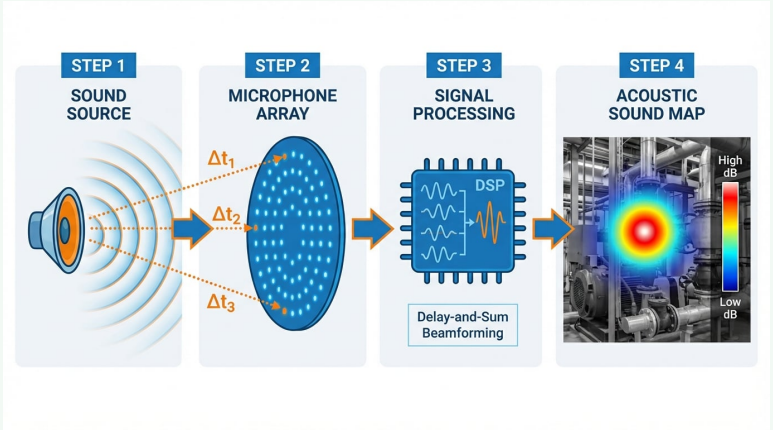
Our Proposed Solution

- The solution we produce eliminates the use of man power as it is completely automated. And we have the concept of 3D mapping which identifies the exact location of the leakage in pipes. In machineries, abnormal frequency of faulty machines is identified at the earliest and reported by the Acoustic camera.

- In higher level the system works by using a network of strategically placed acoustic camera nodes for industries which continuously monitors and process the sound waves and filter out the background noises to distinguish the fault signals. Finally the data from the cameras are reconstructed to form a 3D map of the environment to track the fault.

- The uses recives complete 3D mapping of their environment and the exact fault location in a spatial view.

Technical Approach



The camera contains hundreds of mics which continuously picks up the surrounding sound. This data is then fed to the Signal processing unit which removes the background noise and points out the error frequency. Finally that data is displayed.

Tech Stack

Frontend: python based 3D dashboard

Backend: Python

Database: PostgreSQL

AI/ML, APIs, or Hardware: FFT, spectrogram analysis, Beam forming, ESP32, thermal sensor,Acoustic camera.

Innovation & Uniqueness

- We totally eliminate the need of human fault finding process. hence avoiding the production delay.
- Identifies faults remotely without equipment shutdowns or high-voltage risks. Filter out heavy plant noise by focusing on ultrasonic bands(20-100kHz).Eliminates manual line by line checks through continuous remote monitoring.
- Uses multi-node TDOA ray-casting for depthoutperforming flat 2D handheld acoustic cameras. Automatically maps extra fault coordinates directly onto a factory 3D digital model.

Feasibility & Scalability

- **Feasibility:** Everything needed is low-cost and off the shelf: MEMS ultrasonic mics (analog ones with an external ADC, or digital PDM mics), an STM32 or FPGA for synchronized sampling, a depth camera, a laptop or Raspberry Pi, and Python for FFT and beamforming.

- **Scalability:** The software already separates faults by frequency signature, so adding partial discharge and bearing wear needs new signatures, not a new system. It then expands from factories to power substations, pharma, food and textile plants.

- **Risks:** 1)Wrong identification of pipes which has multiple layer.
  2) Changing the orentation of the camera may lead to error results.

Cost for Product (Optional)

- Development or hardware cost: 1,80,000rs Sensors, microphone array, data acquisition, depth camera, electronics, and prototype assembly.
- Hosting or maintenance cost: 60,000rs/year Cloud hosting, data backups, dashboard upkeep, sensor calibration, and minor repairs.
- Estimated total: 2,40,000rs. (1,80,000rs initial build + 60,000 first-year hosting and maintenance.)

Use Cases

- 1) Industrial gas and pneumatic leaks :Such as compressed air nitrogen or fuel lines are detected by the high frequency ultrasonic turbulence (20–100kHz) present when the gas escapes at a high pressure through micro-cracks or lose fittings. This is found by using MEMS microphone arrays and delay-and-sum beamforming algorithms that cancel out surrounding plant noise and by multi-node Time Difference of Arrival (TDOA) triangulation to find the (X, Y, Z) spatial coordinates of the leak.

- Faults in electrical machines: such as bearing wear, eccentric rotors, stator winding shorts, and high voltage partial discharge is located by abnormal frequencies. This is achieved by recording acoustic the vibration and applying spectral analysis (such as Fast Fourier Transform) to get faulty frequencies from plant operating noise and by using acoustic imaging and triangulation to locate the exact position of the electrical or mechanical failure.
  Note: Additional slides may be included only for this section.

Use Cases

- Wear and tear in mechanical components: such as motor belt slippage, and gear meshing faults are identified by detecting high frequency friction sounds, impact shocks, and acoustic emissions. This is technically achieved by capturing high-frequency acoustic noise generated during mechanical rubbing or micro-impacts, applying spectral or envelope analysis to extract specific frequencies from plant background sounds, and using acoustic imaging to locate the physical part that is degrading.

Impacts and Benefits

- 1) instead of scanning the whole industrial plant manually, by implementing this idea leaks will be detected and pinpoints in 3D environment in minutes not hours, Without any labour.
  2)Every fault gets a severity score and a Rs cost estimate, so managers can prioritise repairs by money lost and justify the spend.

- when a leak is occured machine will run even more harder to cope up with the required output, which will eventually waste energy, money and time. Continuous detection and a soon repair after will help to cut electricity bill, energy and time

- Time and energy saved: Our system cuts inspection time by at least 50percent compared with manual scanning, and estimates the energy and money saved for each leak fixed: leak size $\times$ compressor power $\times$ running hours $\times$ electricity tariff.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 9	Target 9.4	Industrial innovation and infrastructure	Modenizes industrial by retofitting acoustic cameras.
SDG 12	Target 12.2	Responsible consumption and production	Eliminating industrial energy waste and raw material losses by detecting pneumatic pressure leaks.

Individual Member Contribution

Member	Contribution or responsibility
SRI SAILASWARAN A	TEAM LEADER
SURYA V	IDEA PERSON
OMESHVARDHAN T P	IDEA PERSON
SURENDRAN M	IDEA PERSON

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